

COMPUTING THREEFOLD FLIPS IN MACAULAY2

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ABSTRACT. We implement Algorithm 3 (Computing a flip) of the author's paper on the minimal model program in dimension three, and run it on explicit threefolds. Given the coordinate ring of the target X of a flipping contraction, the implementation produces the flip as

$$Z = \text{Proj } R[I^{(m)}t] \longrightarrow X,$$

together with the certificates of Lemma 7.2 — that Z is normal and that the exceptional locus has codimension at least two. The centrepiece is a genuine three-dimensional flip of a non- $\mathbb{Q}$ -Gorenstein toric singularity, computed from end to end and checked against toric geometry derived independently of the algorithm. This note is not a formal verification; it records a reproducible experiment and sets out precisely which claims the computations do and do not support. The code and this note were written essentially by an AI system, with edits by the author; see Section 8.

1. PURPOSE AND BACKGROUND

Let $f : Y \rightarrow X$ be a flipping contraction: a small projective birational morphism of normal threefolds with $-K_Y$ f -ample and relative Picard rank one. Its **flip** is the small projective birational $g : Z \rightarrow X$ with K_Z g -ample. Existence is a theorem; the question here is whether the flip can be *computed* from equations.

Algorithm 3 of the paper answers this by the classical construction

$$Z = \text{Proj}_X \bigoplus_{k \geq 0} \mathcal{O}_X(-kmE), \quad E \sim -K_X,$$

which is a flip once m is chosen so that the algebra is finitely generated and the result is normal and small. The algorithm searches for such an m and verifies the two conditions rather than assuming them.

The input is the homogeneous coordinate ring R of X , so that $X = \text{Proj } R$ sits in a weighted projective space; the output is a **B2M projection**, a bigraded variety $Z \subset \mathbb{P}^{r-1} \times X$ together with its projection to X (Definition 2.4 of the paper). Beyond the scope of the paper, the implementation also accepts an affine base $X = \text{Spec } R$, which is what makes computation at a single singularity cheap.

2. WHAT THE ALGORITHM COMPUTES

Writing $K_X = \sum_i n_i D_i$ with $D_i = V(p_i)$ of height one, the five steps are:

1. compute K_X ;
2. choose a homogeneous s with $-K_X + \text{div}(s)$ effective;
3. form $I = \mathcal{O}_X(K_X - \text{div}(s))$ and E with $I = \mathcal{O}_X(-E)$;
4. for $m = 1!, 2!, 3!, \dots$;
5. build $Z^m = \text{Proj } R[I^{(m)}t]$ and test Lemma 7.2 — return Z^m at the first m that passes.

Steps 1–3 are `flipDivisorData`, step 5 is `bigradedReesProjection` together with `isSmallProjection` and `isS2Source`, and the loop is `computeFlip`.

2.1. Where the implementation departs from the paper. Four departures matter mathematically. The first is a correction; the others are reformulations that leave the answer alone and change the cost by orders of magnitude. [docs/implementation-notes.md](#) gives the full argument and the measurements for each.

Weights on the fibre, not levelled generators. Presenting Z^m inside $X \times \mathbb{P}^{r-1}$, with one fibre coordinate u_i per generator f_i of $I^{(m)}$, makes the Rees relations bihomogeneous only if the f_i share a degree — $u_i \mapsto f_i t$ forces $\deg u_i = \deg f_i + \deg t$ to be independent of i . Replacing $I^{(m)}$ by the ideal generated by its degree- D part is the obvious repair and is what this package originally did. **It is unsound over a weighted grading:** that ideal is $\sum_i R_{D-d_i} f_i$, and R_{D-d_i} can vanish, dropping f_i outright. Grading the threefold below by $\ell = (3, 2, 1)$ gives ring degrees 2, 5, 3, 6, 7 and generators in degrees 2 and 3, so $D = 3$ needs $R_1 = 0$ — and the projection is then reported as *not small when it is small*. Enlarging D does not help; the D that preserve the sheaf are sparse, non-monotone, and sometimes absent. The implementation instead weights the fibre direction, with $d_0 = \min_i d_i$:

$$\deg u_i = (1, d_i - d_0), \quad \deg x_j = (0, c_j), \quad \deg t = (1, -d_0),$$

so that $u^a x^b$ has bidegree $(|a|, \deg(f^a x^b) - |a|d_0)$ — exactly what its image $f^a x^b t^{|a|}$ records. The kernel is bihomogeneous whatever the degrees of the f_i , and when they agree this *is* the paper's $\deg u_i = (1, 0)$.

Normality reduces to S_2 . A small projection satisfies R_1 automatically: a codimension-one point of Z cannot lie in the exceptional locus, so π is a local isomorphism there and the local ring is that of X at a codimension-one point, hence a DVR. Testing smallness *first* therefore reduces normality to S_2 . This is not a small saving: on one example the R_1 half of `isNormal` took 119 s, passing through a Jacobian minor ideal with 1 436 495 generators, against 0.02 s for S_2 .

ω_X embedded directly. Steps 2 and 3 together amount to embedding ω_X into R as an ideal, so instead of choosing s the implementation takes a homomorphism $\omega_X \rightarrow R$ of least degree. The s of Step 2 inherits whichever representative of the canonical class the `WeilDivisors` package returns; under the grading $\ell = (2, 2, 1)$ that gives $s = y_3^{11}$ and an $I^{(1)}$ in degree 30, and the computation does not finish in seventeen minutes. The direct embedding gives generators of degree 2 and takes 0.05 s — on the same threefold.

Every divisor of `MaxSteps!`. One m that works is as good as another, so what matters is finding a *small* one, and the cost per iteration grows steeply: on the threefold below, 0.02 s at $m = 1$ against more than 600 s at $m = 12$. Trying every divisor of `MaxSteps!` in increasing order still contains $1!, 2!, \dots$, so termination is unchanged, while filling the gaps is nearly free.

3. ENVIRONMENT AND REPRODUCIBILITY

Computed with Macaulay2 1.24.11 on macOS (Apple Silicon), using the bundled packages `WeilDivisors`, `SymbolicPowers`, `Polyhedra`, `IntegralClosure`, `MinimalPrimes` and `Elimination`. From the repository root:

```
M2 --script tests/run-tests.m2          # 13 regression tests, about 5 s
M2 --script examples/toric-flip.m2      # about 1 s
```

Measured wall-clock times: the test suite 5 s, `toric-flip.m2` 1.0 s, `toric-flip-projective.m2` 1.3 s, `toric-flip-index-two.m2` 2.7 s, `ordinary-double-point.m2` 0.6 s. The checks quoted below — fans, signs, dimensions, smoothness, the agreement between code paths — are **asserted** by the regression suite rather than merely printed, so a regression shows up as a test failure. A few remarks are derivations made here for the reader rather than assertions in the suite, and are marked as such. The excerpts are trimmed; the complete input and output are in the repository.

The systems of equations quoted below are exactly what this version returns. They are Gröbner basis output, so the order in which the generators are listed — though not the ideal they generate, and not any statement made about it here — may differ under another version of Macaulay2.

4. REPRESENTATIVE MACAULAY2 CALCULATIONS

4.1. A flip of a non- $\mathbb{Q}$ -Gorenstein toric threefold. Let $\sigma \subset N_{\mathbb{R}} = \mathbb{R}^3$ be the cone on

$$v_1 = (1, 0, 0), \quad v_2 = (0, 1, 0), \quad v_3 = (0, 0, 1), \quad v_4 = (1, 1, -2),$$

with circuit relation $v_1 + v_2 = 2v_3 + v_4$, and let $X = \text{Spec } k[\sigma^\vee \cap M]$. No $m \in M_{\mathbb{Q}}$ has $\langle m, v_i \rangle = 1$ for all four rays — the first three force $m = (1, 1, 1)$ and then $\langle m, v_4 \rangle = 0$ — so X is **not $\mathbb{Q}$ -Gorenstein**, and this is the case the paper is aimed at.

The Hilbert basis of $\sigma^\vee \cap M$ has five elements, so this X is a threefold in $\mathbb{A}^5$, and the input handed to the algorithm is the three quadrics cutting it out:

$$y_4^2 - y_2 y_5 \quad y_3 y_4 - y_1 y_5 \quad y_2 y_3 - y_1 y_4$$

equivalently, the 2×2 minors of

$$\begin{pmatrix} y_1 & y_2 & y_4 \\ y_3 & y_4 & y_5 \end{pmatrix}.$$

A circuit has exactly two triangulations, and toric intersection theory gives the sign of $K \cdot C$ on the wall curve from the wall relation alone:

triangulation	wall	$K \cdot C$
$\langle v_1, v_2, v_3 \rangle, \langle v_1, v_2, v_4 \rangle$	$\langle v_1, v_2 \rangle$	$-(2 + 1 - 1 - 1) = -1$
$\langle v_1, v_3, v_4 \rangle, \langle v_2, v_3, v_4 \rangle$	$\langle v_3, v_4 \rangle$	$-(1 + 1 - 2 - 1) = +1$

So the first is the flipping contraction $Y \rightarrow X$ and **the second is the flip** $Z \rightarrow X$. This is known before the algorithm is run, and it is what the output is compared against. The algorithm succeeds at $m = 1$:

```
-- I = omega_X, embedded in R by a map of least degree
-- E has 2 component(s) with multiplicities {1, 1}
-- m = 1:
--   Z^m sits in P^1 x X, dim Z^m = 3
--   component of dim 2 (not contracted)
--   component of dim 2 (not contracted)
-- the flip is obtained for m = 1
```

and returns $Z \subset \mathbb{P}^1 \times X$ cut out by

$$\begin{array}{lll} y_4^2 - y_2 y_5 & y_3 y_4 - y_1 y_5 & u_1 y_4 - u_2 y_5 \\ y_2 y_3 - y_1 y_4 & u_1 y_2 - u_2 y_4 & u_1 y_1 - u_2 y_3 \end{array}$$

The three equations of X survive, and three more appear, each linear in u . Together they are once again the 2×2 minors of a single matrix — the matrix of X with one column adjoined:

$$I_2 \begin{pmatrix} y_1 & y_2 & y_4 \\ y_3 & y_4 & y_5 \end{pmatrix} \rightsquigarrow I_2 \begin{pmatrix} u_2 & y_1 & y_2 & y_4 \\ u_1 & y_3 & y_4 & y_5 \end{pmatrix}.$$

On this example, then, the flip is the operation of adjoining the fibre coordinates as a new column. The identity is not read off by eye: the ideal returned by `computeFlip` is compared with `minors(2,M)` in Macaulay2 and they are equal, and Z has codimension three and degree four.

The output is then dissected:

```

blown up ideal I = 0_X(-E): ideal (y_5, y_4)
fibre over the fixed point has dimension 1
chart u_1 = 1: dim 3, smooth: true
chart u_2 = 1: dim 3, smooth: true

the fan of Z has maximal cones
  {{0, 1, 0}, {1, 1, -2}, {0, 0, 1}}
  {{1, 0, 0}, {1, 1, -2}, {0, 0, 1}}
equals the expected triangulation {v1,v3,v4}, {v2,v3,v4}: true

Z: wall relation {1, 1, -1, -2}, K.C ~ 1
Y: wall relation {1, 2, -1, -1}, K.C ~ -1

```

K_Z is π -ample and K_Y is π -anti-ample, so Z is the flip of Y .

Three things are confirmed at once. I has exactly two generators, so Z lands in $\mathbb{P}^1 \times X$ — one homogeneous coordinate per maximal cone of Z . The fan recovered from those generators is the triangulation $\langle v_1, v_3, v_4 \rangle, \langle v_2, v_3, v_4 \rangle$. And the sign of $K \cdot C$, computed from the wall relation, is $+1$: the algorithm returned the flip and not the contraction it came from.

Also checked here: `canonicalDivisorData` returns $-D_1 - D_3$ rather than the torus-invariant $-\sum_i D_i$, and the two differ by the principal divisor $\text{div}(y_1) = D_2 + D_4$, so they are linearly equivalent as they must be.

4.2. The same flip over a projective base. Every element of the Hilbert basis pairs to 1 with $\ell = (1, 1, -1)$, so the standard grading is induced by a torus character and

$$X_{\text{proj}} = \text{Proj } k[y_1, \dots, y_5, w]/I \subset \mathbb{P}^5, \quad \deg w = 1$$

is a projective threefold with $D_+(w) = \text{Spec } R$. Projecting the rays of $\tau = \sigma \times \mathbb{R}_{\geq 0}$ shows its only singularity is the vertex — the point to be flipped — so this is the paper's setting with nothing extra added.

```

degrees for l = (1,1,-1): {1, 1, 1, 1, 1} (so the standard grading works)
singular locus is the vertex [0:...:0:1]: true
-- the flip is obtained for m = 1
I = ideal (y_5, y_4), uniform degree used = null
affine I was ideal (y_5, y_4), the same ideal
Z_proj: 12 nonempty charts, dimensions {3}, all smooth: true
Z_proj restricted to D_+(w) equals the affine flip: true

```

The last line is the check that matters: the graded and ungraded code paths, which share very little, agree *exactly* on the overlap.

4.3. A flip that needs $m = 2$. Replacing v_4 by $(1, 3, -2)$, with circuit relation $v_1 + 3v_2 = 2v_3 + v_4$, reverses which triangulation carries the π -ample canonical class, and the flip becomes singular: $\langle v_1, v_2, v_4 \rangle$ has determinant -2 , and $\langle m, v_1 \rangle = \langle m, v_2 \rangle = \langle m, v_4 \rangle = 1$ needs $m = (1, 1, 3/2)$, so K_Z is 2-Cartier but not Cartier. This is the first example in which the loop over m actually iterates:

```

-- m = 1:
--   Z^m sits in P^2 x X, dim Z^m = 3
--   component of dim 2 (contracted)
--   component of dim 2 (not contracted)
--   exceptional locus contains a divisor
-- m = 2:
--   Z^m sits in P^1 x X, dim Z^m = 3

```

```

component of dim 2 (not contracted)
-- the flip is obtained for m = 2

I^(2) = ideal (y_6, y_4^2)
the fan of Z ... equals the expected triangulation {v1,v2,v3}, {v1,v2,v4}: true
chart u_1 = 1: dim 3, smooth: false
chart u_2 = 1: dim 3, smooth: true

```

Here the input is six equations rather than three, of degrees 2, 2, 2, 3, 3, 3:

```

y_5^2-y_3*y_6      y_4*y_5-y_2*y_6      y_3*y_4-y_2*y_5
y_2*y_4^2-y_1*y_6  y_2^2*y_4-y_1*y_5      y_2^3-y_1*y_3

```

All six survive in the output, joined by four that are linear in u :

```

u_1*y_1-u_2*y_2      u_1*y_2^2-u_2*y_3
u_1*y_2*y_4-u_2*y_5  u_1*y_4^2-u_2*y_6

```

Nothing as tidy as the determinantal identity above holds here: the 2×2 minors of the corresponding 2×5 matrix lie in the ideal but do not generate it. What these equations do show is the weighted fibre grading at work. The two generators of $I^{(2)} = (y_6, y_4^2)$ have degrees 1 and 2, so they are not levelled; the fibre weights are $\deg u_1 = (1, 0)$ and $\deg u_2 = (1, 1)$, and the last relation $u_1 y_4^2 - u_2 y_6$ is not bilinear. It is bihomogeneous only for those weights.

$m = 1$ is rejected for the right reason — a *contracted* component of dimension two, that is a divisor in the exceptional locus — and not by the normality test, which Z^1 passes. The index of K_Z being 2 and the required m being 2 agree, and one chart is singular exactly as the determinant predicts.

4.4. The ordinary double point, and the graph morphism. The small resolution of the threefold ODP $x_0 x_1 = x_2 x_3$ is the standard sanity check, and it also exercises the Section 2 machinery: the Segre construction of Lemma 2.3 turns the B2M projection into a genuine morphism of monograded varieties (Lemma 2.6).

```

K_X = {{ideal(x2,x0), -3}, {ideal(x3,x0), -3}}
Z is normal:      true
pi is small:      true
-- Segre product has 10 generators
graph morphism with graph of dim 3

```

The canonical divisor is an independent check, though this last reading is a derivation rather than an assertion in the suite: $D_1 = V(x_0, x_2)$ and $D_2 = V(x_0, x_3)$ are the two planes cut by $x_0 = 0$, so $D_1 + D_2 = \text{div}(x_0) \sim H$ and the computed $-3D_1 - 3D_2$ is $-3H$ — the right canonical class for a quadric threefold.

5. WHY THE ANSWERS ARE BELIEVED TO BE CORRECT

No claim of formal verification is made. What the repository does provide is a set of checks that are **independent of the algorithm being tested**, in the sense that they are computed from the toric data or from classical geometry rather than from the object the algorithm returns.

1. The answer is compared with an independently known answer. For a circuit there are exactly two triangulations, and which one is the flip is determined by the sign of $K \cdot C$ read off from the wall relation. The tests recompute that sign from the fan and compare. The algorithm is never asked to certify itself.

2. Flip versus antiflip is checked, not assumed. This is the sharpest test, because both triangulations give a small birational morphism to X and only the sign of $K \cdot C$ separates them. A sign error in Step 3 would return the contraction instead of its flip and would otherwise look entirely healthy. This check is what gives confidence in the sign the implementation uses: since

$E = \text{div}(s) - K_X$ is effective, the ideal of $\mathcal{O}_X$ corresponding to $\mathcal{O}_X(K_X - \text{div}(s)) = \mathcal{O}_X(-E)$ carries the exponent $m_i - n_i$, and the computation confirms that this choice lands on the π -ample side.

3. Independent code paths are made to agree. The affine route uses no grading at all; the projective route grades everything. They share the divisor and Rees code but nothing of the degree bookkeeping. On $D_+(w)$ they produce literally the same ideal. Similarly, the two routes to I — the paper's s and the direct embedding of ω_X — produce the same divisor E , and `isS2Source` agrees with the full `isNormalSource` on every example where the latter is affordable.

4. Negative controls. Tests that must *fail* do fail: blowing up a point of $\mathbb{P}^3$, or the origin of $\mathbb{A}^3$, is divisorial and is rejected by the smallness test; $m = 1$ is rejected in the index-two example. A test suite of positives only would not detect a smallness test that always returns true.

5. Classical invariants. $K_{\mathbb{P}^3} = -4H$; the quadric threefold has $K = -3H$; the Segre product of two copies of $\mathbb{P}^1$ has four generators and $\mathbb{P}^1 \times \mathbb{P}(1, 2)$ has five. These pin down the supporting routines independently of any flip.

6. The checks have demonstrably caught a real error. This is the strongest evidence that they are not vacuous. Levelling the generators of $I^{(m)}$ to a single degree — the natural way to make the presentation in $X \times \mathbb{P}^{r-1}$ work, and what this package did originally — reports the toric flip above as **not small** when the base is graded by $\ell = (3, 2, 1)$, while the affine route reports it as small. Seven of eight mixed-degree cases tried had the wrong saturation. The discrepancy between two routes that must agree is what exposed it; it is now a regression test, and the corrected grading is described above.

5.1. What this does not establish.

- It is not a proof of the algorithm, and not a proof that the implementation matches the paper. It is evidence that on these inputs the two agree.
- All flips computed here are **toric**, and all are settled by $m = 1$ or $m = 2$. Nothing is known about the behaviour at large m beyond the timings.
- Over a weighted grading the S_2 test is *sufficient* but not proved necessary, so the loop may in principle try one more m than it has to. This affects optimality, not correctness.
- No check is made that the input really is the target of a flipping contraction. The algorithm assumes it, and will return something for input that is not.
- The base field is $\mathbb{Q}$ in every example; the paper works over $\overline{\mathbb{Q}}$. Nothing in the code assumes $\mathbb{Q}$, but number fields are untested.

6. SUMMARY OF RESULTS

Example	Base	m	$\dim Z$	Independent check
Toric $(1, 1, -2)$	Spec	1	3	fan = expected triangulation; $K \cdot C = +1$
Toric $(1, 1, -2)$	Proj	1	3	restricts to the affine answer on $D_+(w)$; 12 smooth charts
Toric $(1, 3, -2)$	Spec	2	3	$m = 1$ rejected; K_Z of index 2; one chart singular
Toric, $\ell = (3, 2, 1)$	Proj	1	3	agrees with affine; levelling gives the wrong answer
ODP, small resolution	both	—	3	normal and small; $K = -3H$; Segre model of dimension 3
$\text{Bl}_{pt} \mathbb{P}^3, \text{Bl}_0 \mathbb{A}^3$	both	—	3	correctly rejected as not small

Interpretation. On these inputs the implementation reproduces flips that are known independently, distinguishes them from their ant flips, rejects divisorial contractions, and agrees with itself

across two code paths and two constructions of I . That is evidence that it behaves as intended here. It is neither a proof of the algorithm nor a statement about inputs of a different shape.

7. LIMITATIONS

- Only toric examples have been run end to end, and only in dimension three.
- Only $m \leq 2$ has been needed; larger m is exercised for cost, not for correctness.
- The Segre construction of Lemma 2.3 requires the fibre variables to have bidegree $(d_j, 0)$, so `b2mToGraphMorphism` does not apply once the fibre weights are nontrivial, and raises an error. Generalising it is the main open problem left by the correction described above.
- There is no affine analogue of the graph morphism at all.
- `isS2Source` dominates the cost at large m (5.9 s of 7.0 s at $m = 8$).
- Number fields, and bases that are not $\mathbb{Q}$, are untested.

8. USE OF AI IN THE IMPLEMENTATION

The Macaulay2 code, the tests, and this note were written essentially by an AI system (Claude), with edits by the author. The author has read them and believes them correct, but has not checked every line or every intermediate computation.

Two things nevertheless make large errors unlikely. First, the algorithm is the one in the paper below; little is attempted here that is conceptually new, so there is limited room for the code to be quietly doing something else. Second, and more to the point, the tests do not merely record what the code computes: each example is checked against geometry established independently — the triangulations of a circuit, the sign of $K \cdot C$ from a wall relation, determinants of cones, canonical classes of $\mathbb{P}^3$ and of a quadric threefold — and those checks run automatically. A serious error would have to reproduce all of them by accident.

That this is not an idle argument is shown by the levelling bug in item 6 above: an error that had passed every existing test was found precisely because two routes that must agree did not.

None of this replaces a proof.

9. CONCLUSION

Algorithm 3 runs, and it produces flips of non- $\mathbb{Q}$ -Gorenstein toric threefolds that can be checked against toric geometry — including the case where the flip is singular and the multiplier $m = 1$ genuinely fails. The most useful outcome is not the number of examples but the chain connecting the construction in the paper, the source code, the actual input, and output that is verified against independently known answers. Along the way the implementation turned up one point — the presentation of Z^m in $X \times \mathbb{P}^{r-1}$, and the levelling of generators it appears to require — that needs a hypothesis or a reformulation once the base carries a weighted grading; it is recorded above and in the implementation notes.

10. REFERENCES

- **Yasuda, T.** (2026). An algorithm for the minimal model program in dimension three. arXiv:2603.13703. <https://arxiv.org/abs/2603.13703> (Section 2 and Algorithm 3 of Section 7; this repository implements Algorithm 3.)
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